

B2 Symmetry and Relativity, 2019 — Model Solutions

Question 1 — Worldlines, constant-force motion, and causal structure

Problem. Define proper time, 4-velocity $\mathbf{U}$, 4-acceleration $\mathbf{A}$ for $X^\mu(\tau)$; show $\mathbf{U} \cdot \mathbf{A} = 0$. Constant 3-force $\mathbf{f} = (f_x, 0, 0)$ from rest: find $x(t), \dot{x}(t), \ddot{x}(t)$. Then draw causal diagrams of $(0,0,0,0)$ in: (c) Minkowski + worldline of charged particle in E_x ; (d) boosted axes; (e) compactified $x \equiv x + 2\pi R$ with the same charged worldline; (f) compactified time $t \equiv t + 2\pi T$.

(a) Proper time, 4-velocity, 4-acceleration

Proper time is the invariant interval along a timelike worldline:

$$c^2 d\tau^2 = -dX \cdot dX = c^2 dt^2 - d\mathbf{x}^2.$$

4-velocity is the proper-time derivative of position:

$$U^\mu = \frac{dX^\mu}{d\tau} = \gamma(c, \mathbf{v}).$$

4-acceleration is the proper-time derivative of 4-velocity:

$$A^\mu = \frac{dU^\mu}{d\tau} = \gamma \frac{d}{dt}(\gamma c, \gamma \mathbf{v}).$$

Compute $U \cdot U$ with metric $(-, +, +, +)$:

$$U \cdot U = -\gamma^2 c^2 + \gamma^2 v^2.$$

Factor γ^2 :

$$U \cdot U = \gamma^2(v^2 - c^2).$$

Use $\gamma^2 = 1/(1 - v^2/c^2)$:

$$U \cdot U = -c^2.$$

Differentiate with respect to τ :

$$\frac{d}{d\tau}(U \cdot U) = 2U \cdot A.$$

Set equal to zero since the right-hand side is constant:

$$2U \cdot A = 0.$$

Hence

$$\boxed{U \cdot A = 0.} \tag{1}$$

(b) Constant 3-force from rest

Newton-II in relativistic form, with $\mathbf{p} = \gamma m \mathbf{v}$:

$$\frac{dp_x}{dt} = f_x.$$

Integrate with $p_x(0) = 0$:

$$p_x = f_x t.$$

Relate momentum to velocity via $p_x = \gamma m v_x$:

$$\gamma m v_x = f_x t.$$

Use the identity $\gamma^2 = 1 + (p/mc)^2$:

$$\gamma = \sqrt{1 + (f_x t/mc)^2}.$$

Divide $p_x = \gamma m v_x$ by γm :

$$v_x = \frac{f_x t/m}{\sqrt{1 + (f_x t/mc)^2}}.$$

Define $a \equiv f_x/m$ so $\dot{x}$ becomes:

$$\dot{x} = \frac{at}{\sqrt{1 + (at/c)^2}}.$$

Integrate $\dot{x}$ with $x(0) = 0$:

$$x(t) = \int_0^t \frac{at' dt'}{\sqrt{1 + (at'/c)^2}}.$$

Substitute $w = at'/c$, $dw = (a/c) dt'$:

$$x(t) = \frac{c^2}{a} \int_0^{at/c} \frac{w dw}{\sqrt{1 + w^2}}.$$

Use $d(\sqrt{1 + w^2}) = w dw/\sqrt{1 + w^2}$:

$$x(t) = \frac{c^2}{a} \left[\sqrt{1 + w^2} \right]_0^{at/c}.$$

Evaluate at the limits:

$$x(t) = \frac{c^2}{a} \left[\sqrt{1 + (at/c)^2} - 1 \right].$$

Differentiate $\dot{x} = at/\sqrt{1 + (at/c)^2}$ by the quotient rule:

$$\ddot{x} = \frac{a\sqrt{1 + (at/c)^2} - at \cdot \frac{a^2 t/c^2}{\sqrt{1 + (at/c)^2}}}{1 + (at/c)^2}.$$

Combine numerator over the common surd $\sqrt{1 + (at/c)^2}$:

$$\ddot{x} = \frac{a[1 + (at/c)^2] - a^3 t^2/c^2}{[1 + (at/c)^2]^{3/2}}.$$

Expand the bracket $a[1 + (at/c)^2]$:

$$\ddot{x} = \frac{a + a^3 t^2/c^2 - a^3 t^2/c^2}{[1 + (at/c)^2]^{3/2}}.$$

Cancel the $a^3 t^2/c^2$ terms:

$$\ddot{x} = \frac{a}{[1 + (at/c)^2]^{3/2}}.$$

Hence

$$x(t) = \frac{c^2}{a} \left[\sqrt{1 + (at/c)^2} - 1 \right], \quad \dot{x} = \frac{at}{\sqrt{1 + (at/c)^2}}, \quad \ddot{x} = \frac{a}{[1 + (at/c)^2]^{3/2}}, \quad a = f_x/m. \quad (2)$$

Limits: $at \ll c \Rightarrow x \rightarrow \frac{1}{2}at^2, \dot{x} \rightarrow at; at \gg c \Rightarrow \dot{x} \rightarrow c, \ddot{x} \rightarrow 0$.

(c) Minkowski causal structure with charged worldline

Future light-cone of O : $|x| < ct, t > 0$ (region I, can be affected by O). Past light-cone: $|x| < -ct, t < 0$ (region II, can affect O). Elsewhere (spacelike, $|x| > c|t|$, region III) is causally disconnected.

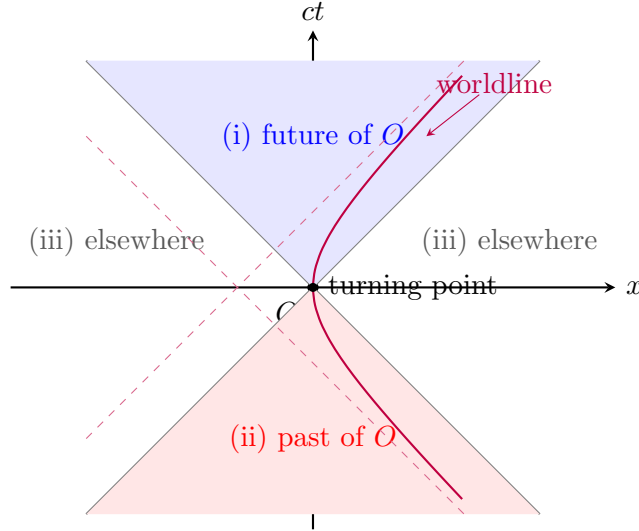
Charged particle from rest in $(E_x, 0, 0)$: $f_x = qE_x$ constant, hyperbolic worldline from (b)

$$x(t) = \frac{c^2}{a} \left[\sqrt{1 + (at/c)^2} - 1 \right], \quad a = qE_x/m.$$

Take $t \rightarrow +\infty$ so $\sqrt{1 + (at/c)^2} \sim at/c$:

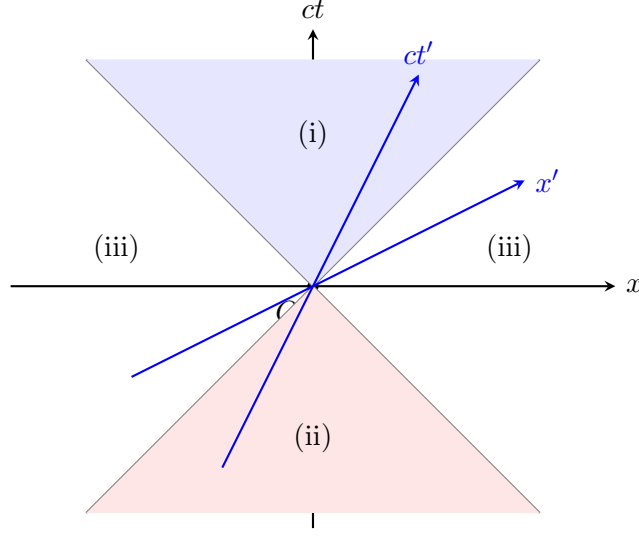
$$x \rightarrow \frac{c^2}{a} \left(\frac{at}{c} - 1 \right) = ct - \frac{c^2}{a}.$$

The worldline asymptotes to the light-cone but never crosses it.



(d) Boosted coordinates

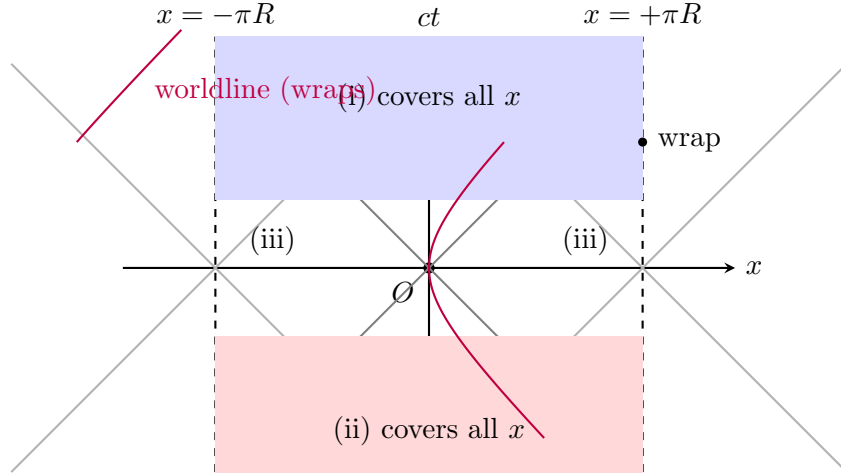
A standard boost rotates both axes inward by the same hyperbolic angle ϕ , $\tanh \phi = v/c$: t' -axis tilts toward the future light-cone, x' -axis tilts symmetrically toward it from the right. The light-cone itself, and therefore the three causal regions (i)/(ii)/(iii), are unchanged.



The light-cone is invariant: causal regions are independent of frame.

(e) Compact spatial dimension $x \equiv x + 2\pi R$

In the universal cover the light-cone is the same, but the identification $x \sim x + 2\pi R$ folds it back. Light from O reaches every point on the spatial circle by $t = \pi R/c$ (going both ways). For $t > \pi R/c$ *every* spatial point lies inside the future cone; similarly the past cone covers all space for $t < -\pi R/c$. The “elsewhere” region is confined to the strip $|t| < \pi R/c$ excluding the local Minkowski cones.



The hyperbolic worldline of (c) wraps around the circle: each time x reaches $+\pi R$ it reappears at $-\pi R$. The particle returns to its starting spatial point with $|\dot{x}|$ larger each time (the hyperbola $\dot{x} = at/\sqrt{1 + (at/c)^2}$ is monotone), approaching c .

Energy conservation. A constant $\mathbf{E}$ -field on a circle requires $\oint \mathbf{E} \cdot d\boldsymbol{\ell} = 2\pi R E_x \neq 0$. By Faraday’s law,

$$\oint \mathbf{E} \cdot d\boldsymbol{\ell} = -\dot{\Phi}_B,$$

so Φ_B through the circle decreases linearly with time at rate $-2\pi R E_x$. The work done on the particle each lap, $qE_x \cdot 2\pi R$, is supplied by the falling magnetic flux. Energy is conserved between particle and field; the assumption of a *static, source-free* constant $\mathbf{E}$ is incompatible with the compactified topology, and a consistent solution requires the dynamical field source.

(f) Compact temporal dimension $t \equiv t + 2\pi T$

Closed timelike curves exist: every event lies on a closed loop $t \rightarrow t + 2\pi T$ with $\mathbf{x}$ fixed. Hence *every* event is to the future and to the past of O simultaneously, so regions (i) and (ii) coincide and cover the whole spacetime; region (iii) is empty.

$$\boxed{\text{(i)} = \text{(ii)} = \text{all events}, \quad \text{(iii)} = \emptyset.} \quad (3)$$

This is inconsistent with causality: the grandfather paradox renders any local field theory ill-posed (no notion of initial data on a Cauchy surface, no global time-ordering). Compact time is therefore physically pathological.

Question 2 — Aberration, transverse Doppler, and resonant absorption by a moving cloud

Problem. (a) Rod of length L along x' moves with $u\hat{\mathbf{y}}'$ in S' ; S' boosted by $v\hat{\mathbf{x}}$ relative to S . Find rod-angle in S . (b) Source at origin emits at f_0 ; observer with velocity $(0, v, 0)$ passes through $(L, 0, 0)$ at $t = 0$: derive $f(t)$ at observer, plot. (c) Replace observer by an unexcited atomic cloud (line f , lifetime τ); plot amplitude and frequency at the source vs t for several v , including the role of f_0/f and intrinsic linewidth.

(a) Apparent angle of the moving rod

Take the rod endpoints simultaneously in S at time t , separated by $\Delta x, \Delta y$ in S . By length contraction along the boost direction:

$$\Delta x = L/\gamma.$$

Apply the inverse Lorentz transform for time, $t' = \gamma(t - vx/c^2)$, at fixed t :

$$\Delta t' = -\gamma v \Delta x / c^2.$$

Substitute $\Delta x = L/\gamma$:

$$\Delta t' = -vL/c^2.$$

In S' both endpoints move with $u\hat{\mathbf{y}}'$, so in this $\Delta t'$ the rightward endpoint shifts by $u\Delta t'$ in y' . Use $\Delta y = \Delta y'$ (no boost in y):

$$\Delta y = u \Delta t' = -\frac{vuL}{c^2}.$$

Form the rod-vector $(\Delta x, \Delta y) = (L/\gamma, -vuL/c^2)$ and compute the angle:

$$\tan \alpha = \frac{\Delta y}{\Delta x} = -\frac{vuL/c^2}{L/\gamma}.$$

Simplify the ratio:

$$\tan \alpha = -\frac{\gamma vu}{c^2}.$$

Hence

$$\boxed{\tan \alpha = -\frac{\gamma vu}{c^2}.} \quad (4)$$

(b) Doppler shift along the observer's path

Observer position in S at time t : $(L, vt, 0)$. Photon emitted from the origin reaches it along

$$\hat{\mathbf{n}}(t) = \frac{(L, vt, 0)}{\sqrt{L^2 + v^2 t^2}}.$$

Define θ as the angle between $\hat{\mathbf{n}}$ and the observer velocity $v\hat{\mathbf{y}}$:

$$\cos \theta = \hat{\mathbf{n}} \cdot \hat{\mathbf{y}} = \frac{vt}{\sqrt{L^2 + v^2 t^2}}.$$

Photon 4-momentum: $P^\mu = (hf_0/c)(1, \hat{\mathbf{n}})$; observer 4-velocity: $U_{\text{obs}}^\mu = \gamma(c, 0, v, 0)$. Observed energy is $-P_\mu U_{\text{obs}}^\mu$:

$$hf_{\text{obs}} = -P_\mu U_{\text{obs}}^\mu = \gamma hf_0(1 - \mathbf{v} \cdot \hat{\mathbf{n}}/c).$$

Divide through by h :

$$f_{\text{obs}} = \gamma f_0(1 - v \cos \theta/c).$$

Substitute $\cos \theta = vt/\sqrt{L^2 + v^2 t^2}$:

$$f_{\text{obs}}(t) = \frac{f_0}{\sqrt{1 - v^2/c^2}} \left(1 - \frac{v^2 t}{c\sqrt{L^2 + v^2 t^2}} \right).$$

Combine the bracket over a common denominator:

$$f_{\text{obs}}(t) = \frac{f_0 (c\sqrt{L^2 + v^2 t^2} - v^2 t)}{c\sqrt{L^2 + v^2 t^2} \sqrt{1 - v^2/c^2}}.$$

Hence

$$\boxed{f_{\text{obs}}(t) = \frac{f_0}{\sqrt{1 - v^2/c^2}} \left(1 - \frac{v^2 t}{c\sqrt{L^2 + v^2 t^2}} \right)}. \quad (5)$$

Take $t \rightarrow -\infty$ so $\cos \theta \rightarrow -1$:

$$f_{\text{obs}} \rightarrow \gamma f_0(1 + v/c).$$

Simplify with $\gamma(1 + v/c) = \sqrt{(1 + v/c)/(1 - v/c)}$:

$$f_{\text{obs}}(-\infty) = f_0 \sqrt{\frac{1 + v/c}{1 - v/c}}.$$

Take $t \rightarrow +\infty$ so $\cos \theta \rightarrow +1$:

$$f_{\text{obs}} \rightarrow \gamma f_0(1 - v/c).$$

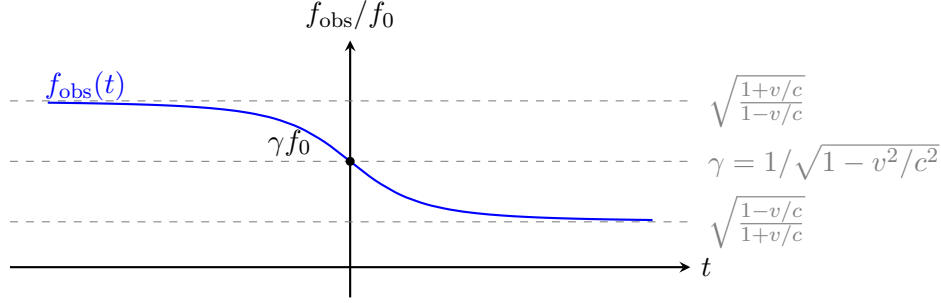
Apply the same identity:

$$f_{\text{obs}}(+\infty) = f_0 \sqrt{\frac{1 - v/c}{1 + v/c}}.$$

At $t = 0$ (transverse passage in S), $\cos \theta = 0$:

$$f_{\text{obs}}(0) = \gamma f_0.$$

The observer approaches from $-y$ (blueshift) and recedes toward $+y$ (redshift); f_{obs} decreases monotonically through the transverse value γf_0 at $t = 0$.



(c) Resonant absorption by a moving atomic cloud and re-emission

Atoms in cloud move with $\mathbf{v} = v\hat{\mathbf{y}}$. Source emits spherical waves at $f_0 < f$. From (5), the frequency seen by an atom is set by $\cos\theta(t) = vt/\sqrt{L^2 + v^2t^2}$:

$$f_{\text{atom}}(t) = \gamma f_0 (1 - v \cos\theta(t)/c).$$

Resonance requires $|f_{\text{atom}} - f| \lesssim \Gamma$, with the natural linewidth $\Gamma \sim 1/(2\pi\tau)$.

Critical velocity v_c . Maximum blueshift occurs at $\cos\theta = -1$ (atom approaching head-on, $t \rightarrow -\infty$):

$$f_{\text{max}}(v) = \gamma f_0 (1 + v/c).$$

Simplify using $\gamma(1 + v/c) = \sqrt{(1 + v/c)/(1 - v/c)}$:

$$f_{\text{max}}(v) = f_0 \sqrt{\frac{1 + v/c}{1 - v/c}}.$$

Resonance possible iff $f_{\text{max}}(v) \geq f$. Setting $f_{\text{max}}(v_c) = f$:

$$\sqrt{\frac{1 + v_c/c}{1 - v_c/c}} = \frac{f}{f_0}.$$

Square both sides and solve for v_c :

$$v_c = c \frac{(f/f_0)^2 - 1}{(f/f_0)^2 + 1}.$$

Resonant angle. Set $f_{\text{atom}} = f$:

$$\gamma f_0 (1 - v \cos\theta_{\text{res}}/c) = f.$$

Solve for $\cos\theta_{\text{res}}$:

$$\cos\theta_{\text{res}} = \frac{c}{v} \left(1 - \frac{f}{\gamma f_0} \right).$$

At $v = v_c$, $f/(\gamma f_0) = 1 + v/c$, giving $\cos\theta_{\text{res}} = -1$ (resonance at head-on). Resonance time t_{res} from $\cos\theta(t) = vt/\sqrt{L^2 + v^2t^2}$:

$$t_{\text{res}} = \frac{L \cos\theta_{\text{res}}}{v \sin\theta_{\text{res}}} = \frac{L \cos\theta_{\text{res}}}{v \sqrt{1 - \cos^2\theta_{\text{res}}}}.$$

For v just above v_c : $\cos\theta_{\text{res}} \rightarrow -1$ so $t_{\text{res}} \rightarrow -\infty$ (resonance with atom far away, approaching head-on). For $v \rightarrow c$: $\gamma \rightarrow \infty$ so $f/(\gamma f_0) \rightarrow 0$, giving $\cos\theta_{\text{res}} \rightarrow 1$ and $t_{\text{res}} \rightarrow +\infty$ (resonance after transverse passage, atom receding nearly head-on).

Frequency of returning photon at source. The excited atom (moving at $v\hat{\mathbf{y}}$ in S) re-emits at rest-frame frequency f ; the photon travels back to the origin with direction $-\hat{\mathbf{n}}$ in S . Contract photon 4-momentum with atom 4-velocity, $-P_\mu U^\mu_{\text{atom}} = hf$:

$$hf = hf_{\text{src}} \gamma(1 + v \cos \theta/c).$$

Solve for the frequency at the source f_{src} :

$$f_{\text{src}} = \frac{f}{\gamma(1 + v \cos \theta/c)}.$$

Rewrite using $1/\gamma = \sqrt{1 - v^2/c^2}$:

$$f_{\text{src}} = \frac{f \sqrt{1 - v^2/c^2}}{1 + v \cos \theta/c}.$$

At the resonance instant use $f/\gamma = f_0(1 - v \cos \theta_{\text{res}}/c)$:

$$f_{\text{src}}^{\text{res}} = f_0 \frac{1 - v \cos \theta_{\text{res}}/c}{1 + v \cos \theta_{\text{res}}/c}.$$

Line width and decay. The transition has finite linewidth $\Gamma \sim 1/(2\pi\tau)$, so resonance is met over a window Δt set by $|df_{\text{atom}}/dt|^{-1}\Gamma$. After absorption, the cloud re-emits exponentially with timescale τ , smearing the return echo by $e^{-(t-t_{\text{res}})/\tau}$ for $t > t_{\text{res}}$.

Three regimes.

(i) $v \ll v_c$: $f_{\text{max}} < f$, no resonance. Absorption only through line wings: amplitude suppression $\propto \Gamma^2/(f - f_{\text{atom}})^2$, exponentially small. Re-emitted signal at source negligible.

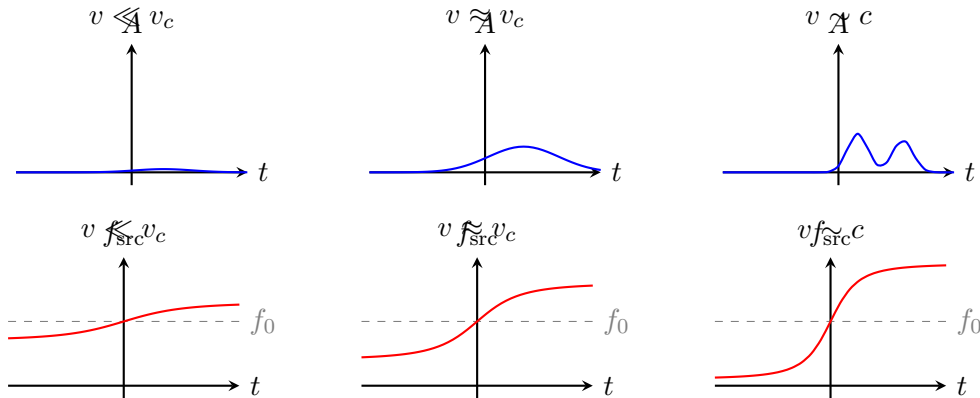
(ii) $v \approx v_c$: resonance only at $\cos \theta \rightarrow -1$, i.e. $t_{\text{res}} \rightarrow -\infty$ (atom far away approaching head-on). Absorption window is broad and asymmetric (vanishing $|df_{\text{atom}}/dt|$ at extremes).

(iii) $v_c < v < c$: a single resonance angle $\cos \theta_{\text{res}} \in (-1, 1)$ gives a sharp absorption notch at t_{res} in the through-going signal, plus a re-emission echo at the source at frequency $f_{\text{src}}^{\text{res}}$, decaying over τ .

(iv) $v \rightarrow c$: $\cos \theta_{\text{res}} \rightarrow 1$, $t_{\text{res}} \rightarrow +\infty$ (resonance occurs late, when the atom is far ahead and receding). Echo arrives at very late time, frequency-shifted strongly by the back-Doppler factor.

Effect of f_0/f . As $f_0/f \rightarrow 1$, $v_c \rightarrow 0$ (resonance for arbitrarily small v); as $f_0/f \rightarrow 0$ (very mismatched), $v_c \rightarrow c$ (only ultra-relativistic clouds resonate).

Schematic plots (v increasing left-to-right). Top: amplitude $A(t)$; bottom: frequency $f_{\text{src}}(t)$ of returning radiation.



Key features: (1) carrier frequency at the source follows the relativistic Doppler curve of (5) (now interpreted as the source-to-atom-to-source two-step); (2) sharp resonant absorption notch whenever $f_{\text{obs}}(t) = f$, of width $\sim \Gamma$; (3) re-emission echo after the absorption notch, decaying with τ , frequency-shifted by the back-Doppler factor; (4) for $v < v_c$, no notches, only line-wing absorption.

Question 3 — 4-vectors, the EM tensor, and field invariants

Problem. (a) Distinguish “invariant” vs “conserved”; how do 4-vectors and tensors transform under Lorentz transformations? (b) Show: if Y^μ has a component zero in all frames, then $Y^\mu = 0$; physical consequence; same for tensors? (c) Form of A_μ and $F_{\mu\nu}$; explicit $F_{\mu\nu}$ in terms of $\mathbf{E}, \mathbf{B}$; $\mathbf{E}, \mathbf{B}$ Lorentz transformations. (d) Show $\mathbf{E}^2 - \mathbf{B}^2$ and $\mathbf{E} \cdot \mathbf{B}$ are Lorentz invariants; show $\int \mathbf{E} \cdot \mathbf{B} d^4x = 0$.

(a) Invariant vs conserved; transformation of tensors

Invariant. A scalar with the same value in every inertial frame. Example: rest mass m , defined by $-P \cdot P = m^2 c^2$; proper time τ ; charge q .

Conserved. A quantity whose total value is constant in time within a chosen frame. Example: total energy or total momentum (both frame-dependent but with $dE/dt = 0$, $d\mathbf{p}/dt = 0$ in an isolated system).

Transformation. A 4-vector transforms as

$$Y'^\mu = \Lambda^\mu{}_\nu Y^\nu,$$

and a rank- (p, q) tensor as

$$T'^{\mu_1 \dots \mu_p}{}_{\nu_1 \dots \nu_q} = \Lambda^{\mu_1}{}_{\alpha_1} \dots \Lambda^{\mu_p}{}_{\alpha_p} (\Lambda^{-1})^{\beta_1}{}_{\nu_1} \dots (\Lambda^{-1})^{\beta_q}{}_{\nu_q} T^{\alpha_1 \dots \alpha_p}{}_{\beta_1 \dots \beta_q}.$$

(b) Vanishing of a component implies vanishing of the 4-vector

Without loss of generality suppose $Y^0 = 0$ in every frame. Apply a general boost with velocity $\mathbf{v}$:

$$Y'^0 = \gamma(Y^0 - \mathbf{v} \cdot \mathbf{Y}/c).$$

Use $Y^0 = 0$:

$$Y'^0 = -\gamma \mathbf{v} \cdot \mathbf{Y}/c.$$

Demand $Y'^0 = 0$ for all $\mathbf{v}$:

$$\mathbf{v} \cdot \mathbf{Y} = 0 \quad \forall \mathbf{v} \implies \mathbf{Y} = 0.$$

Combined with $Y^0 = 0$:

$$\boxed{Y^\mu = 0.} \tag{6}$$

The same argument (boost along any axis) covers vanishing of any single component.

Physical consequence. If a 4-current $J^\mu = (c\rho, \mathbf{j})$ has, say, charge density $\rho = 0$ in every inertial frame, then $\mathbf{j} = 0$ as well: there is no charge-free current. Equivalently, the existence of a current implies the existence of a charge density in some frame.

Tensor counterexample. Take any antisymmetric rank-2 tensor $T^{\mu\nu} = -T^{\nu\mu}$ (e.g. $T^{xy} = 1$, all other independent components zero). Antisymmetry is preserved under Lorentz transformations:

$$T'^{\mu\nu} = \Lambda^\mu{}_\alpha \Lambda^\nu{}_\beta T^{\alpha\beta},$$

$$T'^{\nu\mu} = \Lambda^\nu{}_\alpha \Lambda^\mu{}_\beta T^{\alpha\beta} = -\Lambda^\nu{}_\beta \Lambda^\mu{}_\alpha T^{\beta\alpha} = -T'^{\mu\nu}.$$

Hence $T'^{00} = 0$ in every frame, yet $T^{\mu\nu} \neq 0$. The statement

$$\boxed{\text{fails for tensors (antisymmetric } F_{\mu\nu} \text{ is the natural EM example).}} \tag{7}$$

(c) The 4-potential, field tensor, and field transformations

The 4-potential is $A^\mu = (\phi/c, \mathbf{A})$, with $A_\mu = \eta_{\mu\nu}A^\nu = (-\phi/c, \mathbf{A})$ in the $(-, +, +, +)$ convention. Define the field tensor:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

Recall the 3-field definitions:

$$\mathbf{E} = -\nabla\phi - \partial_t\mathbf{A}, \quad \mathbf{B} = \nabla \times \mathbf{A}.$$

Compute F_{0i} using $\partial_0 = (1/c)\partial_t$ and $A_0 = -\phi/c$:

$$F_{0i} = \partial_0 A_i - \partial_i A_0 = \frac{1}{c}\partial_t A_i + \frac{1}{c}\partial_i \phi.$$

Factor $-1/c$ and identify E_i :

$$F_{0i} = -\frac{1}{c}(-\partial_t A_i - \partial_i \phi) = -E_i/c.$$

Compute the spatial components:

$$F_{ij} = \partial_i A_j - \partial_j A_i = \epsilon_{ijk} B^k.$$

With $c = 1 = \epsilon_0 = \mu_0$:

$$F_{\mu\nu} = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & B_z & -B_y \\ E_y & -B_z & 0 & B_x \\ E_z & B_y & -B_x & 0 \end{pmatrix}. \quad (8)$$

Field transformations. Boost along $\hat{\mathbf{x}}$ with $\beta = v/c$:

$$F'^{\mu\nu} = \Lambda^\mu{}_\alpha \Lambda^\nu{}_\beta F^{\alpha\beta}.$$

Boost matrix:

$$\Lambda^\mu{}_\nu = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Compute F'^{02} ; since $\Lambda^2{}_\beta = \delta^2{}_\beta$ only $\beta = 2$ contributes:

$$F'^{02} = \Lambda^0{}_\alpha F^{\alpha 2}.$$

Only $\alpha = 0, 1$ give non-zero $\Lambda^0{}_\alpha$:

$$F'^{02} = \Lambda^0{}_0 F^{02} + \Lambda^0{}_1 F^{12}.$$

Substitute $F^{02} = E_y$, $F^{12} = B_z$ (units $c = 1$):

$$F'^{02} = \gamma E_y + (-\gamma\beta) B_z.$$

Factor γ :

$$F'^{02} = \gamma(E_y - \beta B_z).$$

Restore c via $\beta = v/c$; repeat the same construction for the remaining entries. The standard result is

$$\boxed{\begin{aligned} E'_x &= E_x, & B'_x &= B_x, \\ E'_y &= \gamma(E_y - vB_z), & B'_y &= \gamma(B_y + vE_z/c^2), \\ E'_z &= \gamma(E_z + vB_y), & B'_z &= \gamma(B_z - vE_y/c^2). \end{aligned}} \quad (9)$$

Compactly $\mathbf{E}'_{\parallel} = \mathbf{E}_{\parallel}$, $\mathbf{B}'_{\parallel} = \mathbf{B}_{\parallel}$, and $\mathbf{E}'_{\perp} = \gamma(\mathbf{E}_{\perp} + \mathbf{v} \times \mathbf{B})$, $\mathbf{B}'_{\perp} = \gamma(\mathbf{B}_{\perp} - \mathbf{v} \times \mathbf{E}/c^2)$.

(d) Invariants and $\int \mathbf{E} \cdot \mathbf{B} d^4x = 0$

Construct two scalars from $F_{\mu\nu}$ and the dual $\tilde{F}^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}F_{\rho\sigma}$. Contract $F_{\mu\nu}\tilde{F}^{\mu\nu}$ component-by-component:

$$\frac{1}{2}F_{\mu\nu}\tilde{F}^{\mu\nu} = \mathbf{B}^2 - \mathbf{E}^2/c^2.$$

Contract $F_{\mu\nu}\tilde{F}^{\mu\nu}$ similarly:

$$\frac{1}{4}F_{\mu\nu}\tilde{F}^{\mu\nu} = -\mathbf{E} \cdot \mathbf{B}/c.$$

Both are full contractions of Lorentz tensors with the invariant $\eta_{\mu\nu}$ and $\epsilon^{\mu\nu\rho\sigma}$, hence Lorentz scalars:

$$\boxed{\mathbf{E}^2 - \mathbf{B}^2 \text{ (up to } c^2) \text{ and } \mathbf{E} \cdot \mathbf{B} \text{ are Lorentz-invariant.}} \quad (10)$$

Vanishing integral. Express $\mathbf{E} \cdot \mathbf{B}$ via the dual (units $c = 1$):

$$\mathbf{E} \cdot \mathbf{B} = -\frac{1}{4}F_{\mu\nu}\tilde{F}^{\mu\nu}.$$

Substitute $\tilde{F}^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}F_{\rho\sigma}$:

$$F_{\mu\nu}\tilde{F}^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}.$$

Sub $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ in the first factor:

$$F_{\mu\nu}\tilde{F}^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}(\partial_\mu A_\nu - \partial_\nu A_\mu)F_{\rho\sigma}.$$

Use antisymmetry of ϵ on $\mu\nu$ to combine the two terms:

$$F_{\mu\nu}\tilde{F}^{\mu\nu} = \epsilon^{\mu\nu\rho\sigma}\partial_\mu A_\nu F_{\rho\sigma}.$$

Apply the product rule to $\partial_\mu(A_\nu F_{\rho\sigma})$:

$$F_{\mu\nu}\tilde{F}^{\mu\nu} = \partial_\mu(\epsilon^{\mu\nu\rho\sigma}A_\nu F_{\rho\sigma}) - \epsilon^{\mu\nu\rho\sigma}A_\nu\partial_\mu F_{\rho\sigma}.$$

Use the Bianchi identity $\epsilon^{\mu\nu\rho\sigma}\partial_\mu F_{\rho\sigma} = 0$:

$$F_{\mu\nu}\tilde{F}^{\mu\nu} = \partial_\mu K^\mu, \quad K^\mu \equiv \epsilon^{\mu\nu\rho\sigma}A_\nu F_{\rho\sigma}.$$

Divide by -4 to obtain $\mathbf{E} \cdot \mathbf{B}$:

$$\mathbf{E} \cdot \mathbf{B} = -\frac{1}{4}\partial_\mu K^\mu.$$

Integrate over spacetime and apply the divergence theorem:

$$\int \partial_\mu K^\mu d^4x = \oint_{\partial} K^\mu d\Sigma_\mu.$$

Boundary term vanishes since $\mathbf{E}, \mathbf{B} \rightarrow 0$ at infinity:

$$\oint_{\partial} K^\mu d\Sigma_\mu = 0.$$

Therefore

$$\boxed{\int \mathbf{E} \cdot \mathbf{B} d^4x = 0.} \quad (11)$$

Question 4 — Klein–Gordon, Yukawa, 2-spinors

Problem. (a) Plane-wave and static spherical solutions of $-\partial_\mu \partial^\mu \phi + m^2 \phi = 0$; restriction on plane waves and physical interpretation. (b) Compare Yukawa potential at 10^{-15} m and 10^{-10} m to its value at 10^{-18} m for $m_\pi = 135$ MeV/ c^2 and $m_H = 125$ GeV/ c^2 ; massless reference; implications for nuclei and molecules. (c) What is a 2-spinor; which particles? (d) $U^\mu = \frac{1}{2} u^\dagger \sigma^\mu u$, $\sigma^\mu = (\mathbf{1}, \boldsymbol{\sigma})$: show U^μ is null; under $u \rightarrow e^{\rho \sigma_z / 2} u$ identify the Lorentz transformation of U^μ ; comment on $u \rightarrow e^{i\theta \sigma_z} u$. (e) Does a 2-spinor satisfy the massless KG equation?

(a) Plane-wave and static spherically symmetric solutions

Use metric $(-, +, +, +)$ so $\partial_\mu \partial^\mu = -\partial_t^2 + \nabla^2$; the equation becomes:

$$\partial_t^2 \phi - \nabla^2 \phi + m^2 \phi = 0.$$

Plane wave. Try $\phi = e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)}$ and compute derivatives:

$$\partial_t^2 \phi = -\omega^2 \phi, \quad \nabla^2 \phi = -k^2 \phi.$$

Substitute into the equation:

$$-\omega^2 \phi + k^2 \phi + m^2 \phi = 0.$$

Divide by ϕ :

$$\boxed{\omega^2 = k^2 + m^2 \quad (\hbar = c = 1).} \quad (12)$$

Restriction: $|\omega| \geq m$. Physical interpretation: a quantum of mass m must carry energy $\geq mc^2$ (rest energy); modes with $\omega < m$ are non-propagating.

Static spherically symmetric. Drop ∂_t :

$$-\nabla^2 \phi + m^2 \phi = 0.$$

In spherical symmetry $\nabla^2 \phi = (1/r) d^2(r\phi)/dr^2$. Set $u(r) = r\phi$:

$$-\frac{1}{r} \frac{d^2 u}{dr^2} + \frac{m^2 u}{r} = 0.$$

Multiply through by $-r$:

$$\frac{d^2 u}{dr^2} = m^2 u.$$

General solution decaying at infinity:

$$u(r) = C e^{-mr}.$$

Divide by r to obtain ϕ :

$$\boxed{\phi(r) = \frac{C e^{-mr}}{r} \quad (\text{Yukawa}).} \quad (13)$$

(b) Yukawa ratio at nuclear/atomic vs 10^{-18} m scales

Write the Compton wavelength

$$\lambda_m = \hbar/(mc).$$

Compute λ_π :

$$\lambda_\pi = \frac{\hbar c}{m_\pi c^2} = \frac{197 \text{ MeV fm}}{135 \text{ MeV}} = 1.46 \text{ fm} = 1.46 \times 10^{-15} \text{ m}.$$

Compute λ_H :

$$\lambda_H = \frac{197 \text{ MeV fm}}{1.25 \times 10^5 \text{ MeV}} = 1.58 \times 10^{-3} \text{ fm} = 1.58 \times 10^{-18} \text{ m}.$$

Yukawa ratio between r_2 and reference r_1 :

$$\frac{\phi(r_2)}{\phi(r_1)} = \frac{r_1}{r_2} e^{-(r_2-r_1)/\lambda_m}.$$

Reference $r_1 = 10^{-18} \text{ m}$.

Pion ($\lambda_\pi = 1.46 \times 10^{-15} \text{ m}$). Nuclear $r_2 = 10^{-15} \text{ m}$:

$$\frac{\phi(10^{-15})}{\phi(10^{-18})} = 10^{-3} \cdot e^{-(10^{-15}-10^{-18})/1.46 \times 10^{-15}} = 10^{-3} \cdot e^{-0.685} \approx 10^{-3} \cdot 0.504 \approx 5 \times 10^{-4}.$$

Atomic $r_2 = 10^{-10} \text{ m}$:

$$\frac{\phi(10^{-10})}{\phi(10^{-18})} = 10^{-8} \cdot e^{-10^{-10}/1.46 \times 10^{-15}} = 10^{-8} \cdot e^{-6.85 \times 10^4} \approx 0.$$

Higgs ($\lambda_H = 1.58 \times 10^{-18} \text{ m}$). Nuclear:

$$\frac{\phi(10^{-15})}{\phi(10^{-18})} = 10^{-3} \cdot e^{-10^{-15}/1.58 \times 10^{-18}} \approx 10^{-3} \cdot e^{-633} \approx 0.$$

Atomic: even more strongly suppressed.

Massless scalar. $\phi \propto 1/r$, so the ratio is purely r_1/r_2 :

$$\frac{\phi(10^{-15})}{\phi(10^{-18})} = 10^{-3}, \quad \frac{\phi(10^{-10})}{\phi(10^{-18})} = 10^{-8}.$$

Implications. (i) Nuclei: pion-mediated (Yukawa) force has $\lambda_\pi \sim 1 \text{ fm}$, i.e. comparable to nuclear separations: the strong force is short-range but *just* reaches across a nucleus, holding it bound. The Higgs, with $\lambda_H \sim 10^{-3} \text{ fm}$, contributes essentially nothing to nuclear binding. (ii) Molecules: at 10^{-10} m , pion- and Higgs-mediated forces are utterly negligible; only massless mediators (photons) survive, hence chemistry is electromagnetic.

(c) 2-spinors

A 2-spinor is a 2-component complex object $u = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$ that transforms under the fundamental $(\frac{1}{2}, 0)$ representation of $SL(2, \mathbb{C})$ (the universal cover of the Lorentz group). It describes *spin- $\frac{1}{2}$ fermions*: in particular massless Weyl fermions (left- or right-handed), and the building blocks of Dirac fermions like the electron and quarks.

(d) Null property and Lorentz transformation of $U^\mu = \frac{1}{2} u^\dagger \sigma^\mu u$

Null. Let $u = (a, b)^T$. Compute U^0 using $\sigma^0 = \mathbf{1}$:

$$U^0 = \frac{1}{2} u^\dagger u = \frac{1}{2} (|a|^2 + |b|^2).$$

Compute U^z with $\sigma_z u = (a, -b)^T$:

$$U^z = \frac{1}{2} u^\dagger \sigma_z u = \frac{1}{2} (|a|^2 - |b|^2).$$

Compute U^x with $\sigma_x u = (b, a)^T$:

$$U^x = \frac{1}{2}(a^*b + ab^*) = \text{Re}(a^*b).$$

Compute U^y with $\sigma_y u = (-ib, ia)^T$:

$$U^y = \frac{1}{2}(-ia^*b + iab^*) = \frac{i}{2}(ab^* - a^*b).$$

Identify $ab^* - a^*b = 2i \text{Im}(ab^*) = -2i \text{Im}(a^*b)$:

$$U^y = \text{Im}(a^*b).$$

Compute the Minkowski norm with $(-, +, +, +)$ (so $U_\mu U^\mu = -(U^0)^2 + |\mathbf{U}|^2$):

$$-(U^0)^2 + (U^x)^2 + (U^y)^2 + (U^z)^2 = -\frac{1}{4}(|a|^2 + |b|^2)^2 + |a^*b|^2 + \frac{1}{4}(|a|^2 - |b|^2)^2.$$

Use $|a^*b|^2 = |a|^2|b|^2$ and combine:

$$U_\mu U^\mu = -\frac{1}{4}[(|a|^2 + |b|^2)^2 - (|a|^2 - |b|^2)^2] + |a|^2|b|^2.$$

Apply $(p+q)^2 - (p-q)^2 = 4pq$:

$$U_\mu U^\mu = -\frac{1}{4} \cdot 4|a|^2|b|^2 + |a|^2|b|^2.$$

Cancel:

$$U_\mu U^\mu = 0.$$

Hence

$$\boxed{U_\mu U^\mu = 0, \quad U^\mu \text{ is null.}} \quad (14)$$

Boost. Expand $e^{\rho\sigma_z/2}$ using $\sigma_z^2 = \mathbf{1}$:

$$e^{\rho\sigma_z/2} = \cosh(\rho/2) \mathbf{1} + \sinh(\rho/2) \sigma_z.$$

Act on $u = (a, b)^T$:

$$u' = \begin{pmatrix} e^{\rho/2}a \\ e^{-\rho/2}b \end{pmatrix}.$$

Compute U'^0 with $|a'|^2 = e^\rho|a|^2$, $|b'|^2 = e^{-\rho}|b|^2$:

$$U'^0 = \frac{1}{2}(e^\rho|a|^2 + e^{-\rho}|b|^2).$$

Compute U'^z similarly:

$$U'^z = \frac{1}{2}(e^\rho|a|^2 - e^{-\rho}|b|^2).$$

Substitute $e^{\pm\rho} = \cosh \rho \pm \sinh \rho$ into U'^0 :

$$U'^0 = \cosh \rho \cdot \frac{1}{2}(|a|^2 + |b|^2) + \sinh \rho \cdot \frac{1}{2}(|a|^2 - |b|^2).$$

Recognise U^0 and U^z :

$$U'^0 = \cosh \rho U^0 + \sinh \rho U^z.$$

Same substitution for U'^z :

$$U'^z = \sinh \rho U^0 + \cosh \rho U^z.$$

Transverse phases cancel: $a'^*b' = (e^{\rho/2}a)^*(e^{-\rho/2}b) = a^*b$. Hence

$$U'^x = U^x, \quad U'^y = U^y.$$

This is a Lorentz boost along $\hat{z}$ with rapidity ρ :

$$\boxed{u \rightarrow e^{\rho\sigma_z/2}u \iff U^\mu \rightarrow \Lambda^\mu{}_\nu(\hat{z}, \rho) U^\nu, \quad \tanh \rho = v/c.} \quad (15)$$

Phase rotation. Expand $e^{i\theta\sigma_z}$ using $\sigma_z^2 = \mathbf{1}$:

$$e^{i\theta\sigma_z} = \cos \theta \mathbf{1} + i \sin \theta \sigma_z.$$

Act on $u = (a, b)^T$:

$$u' = \begin{pmatrix} e^{i\theta}a \\ e^{-i\theta}b \end{pmatrix}.$$

Moduli unchanged so U^0 and U^z are unchanged:

$$U'^0 = U^0, \quad U'^z = U^z.$$

Compute the cross product a'^*b' :

$$a'^*b' = e^{-i\theta}a^* \cdot e^{-i\theta}b = e^{-2i\theta}a^*b.$$

Write $a^*b = U^x + iU^y$ and expand $e^{-2i\theta}$:

$$a'^*b' = (\cos 2\theta - i \sin 2\theta)(U^x + iU^y).$$

Take real and imaginary parts:

$$U'^x = \text{Re}(a'^*b') = \cos(2\theta)U^x + \sin(2\theta)U^y,$$

$$U'^y = \text{Im}(a'^*b') = -\sin(2\theta)U^x + \cos(2\theta)U^y.$$

This is a spatial rotation by angle -2θ about $+\hat{z}$ (equivalently, 2θ about $-\hat{z}$); the factor of two reflects $SU(2) \rightarrow SO(3)$ being a double cover. Hence

$$\boxed{u \rightarrow e^{i\theta\sigma_z}u \iff U^\mu \rightarrow R^\mu{}_\nu(\hat{z}, -2\theta) U^\nu.} \quad (16)$$

(e) Does a 2-spinor satisfy the massless KG equation?

The bilinear $U^\mu = \frac{1}{2}u^\dagger\sigma^\mu u$ is null, $U_\mu U^\mu = 0$, which is the dispersion $\omega^2 = k^2$ of a massless field. Try a plane-wave 2-spinor $u(x) = u_0 e^{ik \cdot x}$ with $k_\mu k^\mu = 0$:

$$\partial_\mu u = ik_\mu u.$$

Apply $\partial^\mu \partial_\mu$ to each component:

$$\partial^\mu \partial_\mu u_a = -k^\mu k_\mu u_a = 0.$$

Equivalently, the Weyl equation $\bar{\sigma}^\mu \partial_\mu u = 0$ (or $\sigma^\mu \partial_\mu u = 0$) squared via $\sigma^\mu \bar{\sigma}^\nu + \sigma^\nu \bar{\sigma}^\mu = 2\eta^{\mu\nu}$ gives:

$$\partial^\mu \partial_\mu u = 0.$$

Hence

$$\boxed{\text{Yes: each component of a massless 2-spinor solution satisfies the massless KG equation.}} \quad (17)$$